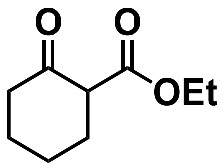
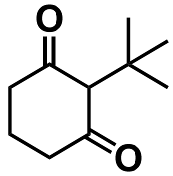
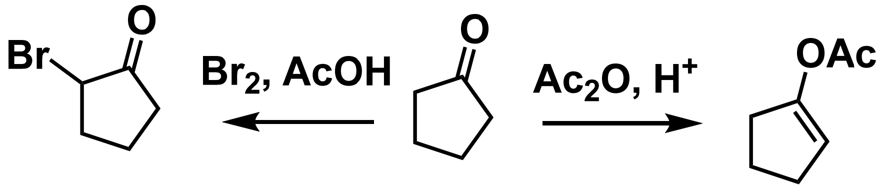
1. Explain the difference in enol proportion for the following two compounds in neat samples.



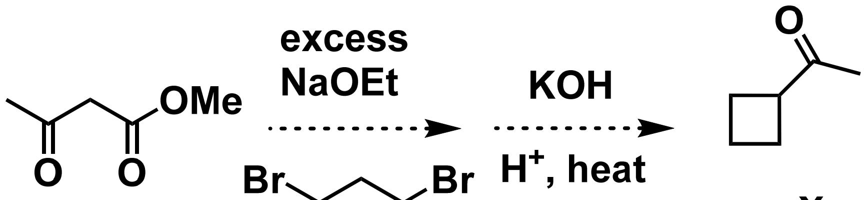



**62% enol** **0% enol**

2. Suggest mechanisms for these reactions and explain why these products are formed.

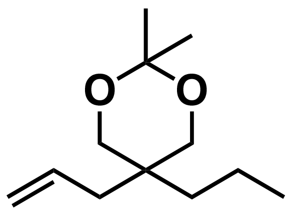



3. Prof. Lin tried to use the following sequence to synthesize Compound X. Unfortunately, he obtained Compound Y instead. After analysis, he found that Y does not contain any ring but does have a carbonyl group. Draw the structure of Y and explain why Prof. Lin did not obtain Compound X.

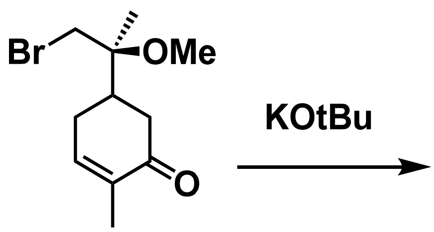



**X**

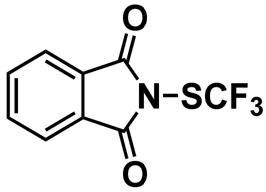
4. Using organic reagents that contain three carbon atoms or less and necessary inorganic reagents, suggest a synthetic plan for the following compound:



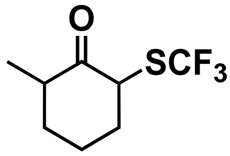
5. Draw the major product for the following reaction and suggest a mechanism.



6. *N*-(Trifluoromethylthio)phthalimide (A) is an electrophilic reagent for trifluoromethylthiolation. Using this reagent and starting from cyclohexanone, suggest a synthetic plan for Compound B.

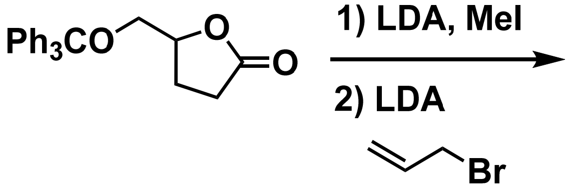


A



B

7. Draw the major product for the following reaction and explain the stereochemistry.




8. Draw the major product for the following reaction and explain the stereochemistry.